

VPN Projects Portfolio

6 Real-World Network Engineering Projects

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Projects	6 VPN Types Covered
Topics	WireGuard · Xray/V2Ray · Enterprise · OpenWrt · GNS3
Level	Intermediate to Expert

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Reverse-Engineered & Rebuilt a Broken VPN Configuration

OVERVIEW

Diagnosed and fully reconstructed a non-functional VPN setup by analyzing logs, misconfigurations, and routing issues. Delivered a clean, working configuration from scratch.

SKILLS

Log Analysis • VPN Debugging • Config
Reconstruction • IPsec/SSL Diagnosis

TOOLS & TECHNOLOGIES

Wireshark • tcpdump • OpenVPN • StrongSwan •
iptables

IMPLEMENTATION STEPS

- 1 Collected existing config files and connection logs
- 2 Identified root causes: certificate mismatch, wrong cipher suites, broken routing
- 3 Rebuilt IPsec/SSL configuration from ground up
- 4 Verified connectivity and tested failover scenarios
- 5 Documented all changes with before/after comparison

OUTCOME: Fully restored VPN connectivity with zero packet loss and proper encryption.

WireGuard VPN with Split Tunneling for 5 Users (Windows & Mac)

OVERVIEW

Deployed a WireGuard VPN server with per-user split tunneling, allowing clients to route only specific traffic through the VPN while keeping local traffic direct.

SKILLS

WireGuard • Split Tunneling • Cross-platform Config
• Key Management

TOOLS & TECHNOLOGIES

WireGuard • wg-quick • Windows WireGuard Client •
Tunnelblick (Mac)

IMPLEMENTATION STEPS

- 1 Provisioned Linux server and installed WireGuard
- 2 Generated key pairs for server and all 5 users
- 3 Configured AllowedIPs for split tunnel per client
- 4 Created Windows (.conf) and Mac profiles
- 5 Tested selective routing and confirmed no DNS leaks

OUTCOME: 5 users connected simultaneously with custom split tunneling rules on both platforms.

Xray/V2Ray VPN — VLESS + Reality Protocol

OVERVIEW

Set up a cutting-edge Xray server using VLESS protocol with Reality camouflage, making VPN traffic indistinguishable from normal HTTPS traffic to bypass deep packet inspection (DPI).

SKILLS

Xray-core • V2Ray • VLESS Protocol • TLS
Fingerprinting • DPI Bypass

TOOLS & TECHNOLOGIES

Xray-core • V2RayN • Nekoray • Nginx • Certbot

IMPLEMENTATION STEPS

- 1 Installed Xray-core on VPS with domain pointed
- 2 Generated Reality keys (private key + short ID)
- 3 Configured VLESS inbound with Reality TLS settings
- 4 Set destination to a real HTTPS site for camouflage
- 5 Connected clients via V2RayN and Nekoray, verified DPI bypass

OUTCOME: Stable, undetectable VPN connection passing through strict firewall environments.

Enterprise VPN — China-to-US Geo-Bypass for Remote Teams

OVERVIEW

Designed and deployed a multi-server VPN architecture enabling remote employees in China to reliably access US-based corporate resources, bypassing the Great Firewall.

SKILLS

Enterprise VPN • GFW Bypass • Multi-hop Routing •
Team Onboarding

TOOLS & TECHNOLOGIES

Xray • WireGuard • HAProxy • Cloudflare CDN •
Ansible

IMPLEMENTATION STEPS

- 1 Analyzed GFW blocking patterns and selected bypass method
- 2 Set up relay server in HK + endpoint server in US
- 3 Configured multi-hop: China -> HK relay -> US destination
- 4 Deployed HAProxy for load balancing across team
- 5 Created onboarding guide and auto-config scripts for all users

OUTCOME: 20+ remote employees in China with stable, fast access to US corporate systems.

Firewall | VPN | Router Configuration — OpenWrt, LAN & WAN

OVERVIEW

Configured an OpenWrt-based router with integrated VPN, custom firewall rules, and proper LAN/WAN segmentation for a secure home or small office network.

SKILLS

OpenWrt • Firewall Rules • VPN on Router •
LAN/WAN Segmentation • iptables/nftables

TOOLS & TECHNOLOGIES

OpenWrt • LuCI • WireGuard (openwrt) • iptables •
dnsmasq

IMPLEMENTATION STEPS

- 1 Flashed OpenWrt firmware on compatible router
- 2 Configured WAN interface and LAN DHCP/DNS
- 3 Installed and configured WireGuard as router-level VPN
- 4 Set up firewall zones: LAN, WAN, VPN with strict rules
- 5 Tested traffic flow, DNS leak protection, and kill switch

OUTCOME: All devices on the network automatically protected by VPN with no per-device setup.

University Network Security Lab — GNS3, IDS/IPS, VPNs & Firewalls

OVERVIEW

Built a complete virtual network security lab in GNS3 simulating enterprise topology with VPNs, intrusion detection/prevention systems, and multi-zone firewalls for academic or training purposes.

SKILLS

GNS3 • IDS/IPS • Snort/Suricata • IPsec VPN •
pfSense • Network Simulation

TOOLS & TECHNOLOGIES

GNS3 • pfSense • Snort • Suricata • Cisco IOS •
OpenVPN

IMPLEMENTATION STEPS

- 1 Designed multi-zone topology: DMZ, LAN, WAN in GNS3
- 2 Deployed pfSense as main firewall with zone-based rules
- 3 Configured site-to-site IPsec VPN between two branches
- 4 Installed Snort/Suricata IDS/IPS with custom rule sets
- 5 Ran attack simulations and verified detection/blocking

OUTCOME: Fully functional security lab used for training, demonstration, and academic research.